

Firmware Lab 4

Design of an hystheretic temperature control

May 25th, 2026

WARNING!!!

**INSTRUCTIONS MUST BE CAREFULLY
READ BEFORE PERFORMING ANY
OPERATION**

Lab goals

The current lab has the goal to implement a simple temeperature control system, based on an hystheretic strategy.

This task is accomplished through the following materials:

- The project template of previous laboratory, already modified by student to acquire temperature and display it on on LCD.
- A VirtLAB board, on which previous firmware is loaded and run.
- A Digital Storage Oscilloscope (DSO), used to perform real word measurements on the VirtLAB board.
- An NTC resistor, and three 10k Ω fixed resistors.

- An NPN transistor, part number 2N2222, used as power switch for a heater, and three resistances, used to drive the switch and to generate heat.
- A breadboard board, to connect electronic components to the VirtLAB.

To correctly accomplish the required goals, a full reading of this documents is mandatory **before** starting to operate on the PC.

Hardware configuration

Temperature measurements are obtained reading the resistance variation of a NTC resistor. This component has a value which varies with the temperature. It has a $10\text{k}\Omega$ resistance at 25°C . The value decreases by 4% for each increase of 1°C of the component temperature.

This subcircuit is the same used in the previous experience, and must be used in the same way.

The two resistive dividers, NTC and fixed one, must be assembled on a breadboard external to the VirtLAB board, and connected through the provided plug/socket cables to the analog port, located on J801. On this port, analog inputs, 3V3 and GND connections are present. The resulting schematic is shown in figure 1.

The heater is built-up by a 47Ω resistor. Attaching it to the power supply, the resistor heat-up, due to power dissipation.

The circuit used to drive the heater is represented in figure 2 and must be assembled on the breadboard, as well. Power and ground connections can be shared with the NTC circuit ones.

As resistors with different values are used in this experience, the student must be able to distinguish them based on the colour code printed on the body of the resistors. A guide to color coding is available at <https://led.polito.it/utilities/utilities.asp#res>.

Firmware structure

The firmware must be derived from the template used in the previous lab. The student must add a task, and the related queue, used to drive the low-side switch, using a GPIO at will.

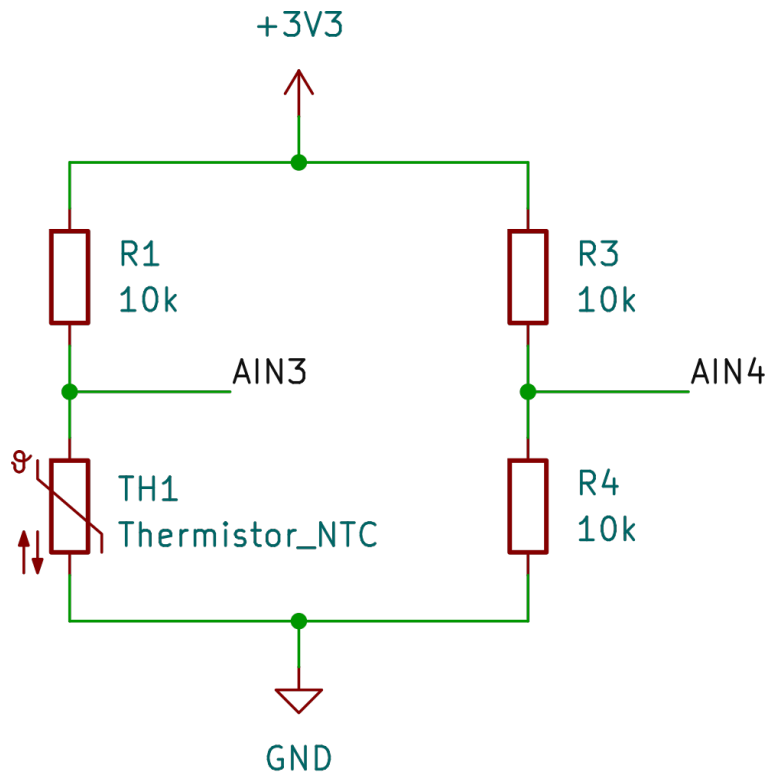


Figure 1: Schematic of dividers and connections to the VirtLAB board, connector J801. AIN3 and AIN4 are the inputs to the ADC converters

The NTC and the heater must be in contact, and the new task must turn on the external transistor if the measured temperature is less than a target temperature. Otherwise, the external transistor must be turned off. The target temperature is 50°C.

Lab report content

The user must provide a detailed description of the source code structure, describing operations executed by every task, including a flowchart for each process. The commented C source code must be attached, too.

The student must also answer to the following questions:

- Which is the calculated dissipated power by the heater?

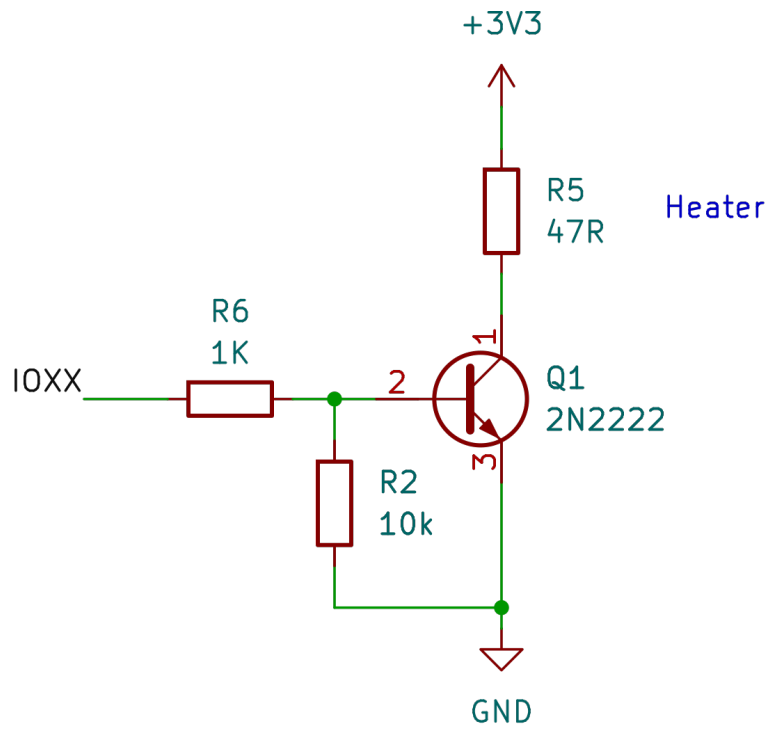


Figure 2: Schematic of the low-side switch and connections to the VirtLAB board.

- Which is the calculated value of the current flowing through the heater?
- Which is the measured value of the current flowing through the heater, and the computed power?